Cafeteria Management System

Technical Documentation

Version 1.0

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Technologies: C# · WPF · SQL Server · ADO.NET

GitHub: github.com/joewaleed/Cafeteria_Project

Date: April 2026

1. Project Overview

The Cafeteria Management System is a desktop application built with C# and WPF (Windows Presentation Foundation) that provides a complete point-of-sale and back-office solution for cafeteria operations. It is backed by a SQL Server database accessed through ADO.NET, offering fast and reliable data handling.

The system supports two user roles (Admin and Cashier) each with a tailored interface and set of permissions. Admins can manage users, items, and categories, while cashiers handle day-to-day billing and order processing.

Property	Details
Project Name	Cafeteria Management System
Platform	Windows Desktop (Windows Forms)
Language	C# (.NET)
Database	SQL Server
Data Access	ADO.NET
GitHub Repository	github.com/joewaleed/Cafeteria_Project
Version	1.0
Author	Yousef Waleed

2. System Architecture

2.1 Technology Stack

Layer	Technology	Purpose
Presentation	Windows Forms	Desktop UI (forms, data grids, navigation)
Business Logic	C# classes	Validation, role checks, cart calculations
Data Access	ADO.NET	SqlConnection, SqlCommand, SqlDataAdapter
Database	SQL Server	Persistent storage for all entities
Security	Password Hashing	Encrypted passwords stored in DB

2.2 User Roles

Feature	Admin	Cashier
Login	✓	✓
Billing / Point of Sale	✓	✓
View Invoices	✓	✓
Manage Users	✓	✗
Manage Items	✓	✗
Manage Categories	✓	✗

3. Database Design

The database consists of four core tables. All tables use integer primary keys with IDENTITY auto-increment. Foreign key relationships enforce referential integrity between Items, Categories, and Invoices.

3.1 Users Table

Column	Type	Description
UserID	INT (PK, IDENTITY)	Unique user identifier
UserName	NVARCHAR(100)	Full name of the user
UserGender	NVARCHAR(10)	Gender (male / female)
UserPass	NVARCHAR(255)	Hashed password
UserPhone	NVARCHAR(20)	Contact phone number
UserAddress	NVARCHAR(255)	Home or work address
IsAdmin	BIT	1 = Admin, 0 = Cashier

3.2 Categories Table

Column	Type	Description
CatCode	INT (PK, IDENTITY)	Unique category identifier
CatName	NVARCHAR(100)	Category display name
CatDesc	NVARCHAR(255)	Short description of the category

3.3 Items Table

Column	Type	Description
ItemCode	INT (PK, IDENTITY)	Unique item identifier
ItemName	NVARCHAR(150)	Product name shown in UI
ItemPrice	DECIMAL(10,2)	Unit price of the item
CatCode	INT (FK → Categories)	Linked category

3.4 Invoices Table

Column	Type	Description
InvoiceID	INT (PK, IDENTITY)	Unique invoice identifier
UserName	NVARCHAR(100)	Cashier who processed the order
InvoiceDate	DATE	Date the order was completed
InvoiceAmount	DECIMAL(10,2)	Total value of the invoice
ItemName	NVARCHAR(150)	Name of each line-item
ItemPrice	DECIMAL(10,2)	Unit price at time of sale
ItemQuantity	INT	Quantity ordered for that item
Total	DECIMAL(10,2)	Line total (price × quantity)

4. Application Screens

The application is composed of six main screens. Each screen is a WPF UserControl loaded dynamically into the main window based on navigation actions.

4.1 Login Page (صفحة دخول)

The entry point of the application. Users enter their username and password. Credentials are validated against the Users table with hashed password comparison. On success, the system determines the role (Admin or Cashier) and loads the appropriate main window.

- Input validation prevents empty submissions.
- Incorrect credentials show a red error label.
- Passwords are never stored or transmitted in plain text.

4.2 User Management (صفحه اداره المستخدمين)

Accessible by Admins only. Displays a DataGrid listing all registered users with columns for ID, name, gender, hashed password, phone, and address. A form on the left allows adding, editing, or deleting users.

- Add User: fill all fields and click Add User.
- Edit User: click a row to populate the form, then click Edit User.
- Delete User: select a row and click Delete User.
- The IsAdmin checkbox controls the role assigned to the user.

4.3 Item Management (صفحة تعديل الاصناف)

Allows the admin to view and maintain the full product catalog. A DataGrid shows ItemCode, ItemName, ItemPrice, and CatName. A form panel handles CRUD operations and category assignment via a dropdown.

- Items are linked to categories through a foreign key (CatCode).
- The Category dropdown is populated dynamically from the Categories table.
- Prices are stored as DECIMAL to ensure accurate financial calculations.

4.4 Category Management (صفحة تعديل التصنيفات)

Manages the logical groupings of menu items. Default categories include Hot Drinks, Cold Drinks, Bakery, and Sandwiches. Each category has a name and a short description.

- Adding a category immediately makes it available in the Items dropdown.
- Deleting a category is restricted if items are linked to it (referential integrity).

4.5 Billing / Point of Sale (صفحة الفواتير و نقطه البيع)

The primary operational screen for cashiers. Provides a two-panel layout: a Categories/Items browser on the left, and a live Cart on the right.

- Clicking a category filters the Items list to that group.
- Selecting an item and entering a quantity then clicking Add to Cart appends it to the cart.
- The cart shows ItemName, Price, Quantity, and Line Total in real time.
- The Total field at the bottom updates immediately as items are added or removed.
- Remove Item removes the selected cart row; Clear Cart empties the entire cart.
- Complete Order saves the invoice to the database and resets the cart.

4.6 Invoice History (صفحه اداره الفواتير)

Provides a searchable log of all past transactions. A lower DataGrid lists invoices (ID, username, date, total). Clicking an invoice row populates an upper DataGrid with the individual line items.

- Search by Order Number: filter by exact invoice ID.
- Search by Date: filter all invoices on a given date.
- Search by Price Range: filter invoices within a minimum/maximum amount.
- The Reset button clears all filters and reloads the full history.

5. Key Features

Feature	Description
Secure Login	Hashed password authentication with role-based access control
Role-based UI	Admin sees full management sidebar; Cashier sees Billing and Invoices only
Real-time Cart	Instant total calculation as items are added, removed, or quantity changes
Invoice Filtering	Search past orders by ID, date, or price range
Password Encryption	Passwords stored as encrypted hashes, never in plain text
Modular Navigation	Sidebar links load UserControls dynamically without restarting windows
Category-based Browsing	Items grid filters by category to speed up the ordering process
Full CRUD	Create, read, update, delete for Users, Items, and Categories

6. Setup & Installation

6.1 Prerequisites

- Windows 10 or later
- Visual Studio 2022 (Community or higher)
- .NET Framework 4.8 or .NET 6+
- SQL Server 2019 or SQL Server Express
- SQL Server Management Studio (SSMS) (optional but recommended)

6.2 Database Setup

- Open SSMS and connect to your SQL Server instance.
- Create a new database named CafeteriaDB.
- Run the provided SQL script (schema.sql) to create all four tables.
- Optionally run seed.sql to insert sample users, categories, and items.

6.3 Application Configuration

- Clone the repository: git clone https://github.com/joewaleed/Cafeteria_Project
- Open the .sln file in Visual Studio.
- In App.config (or appsettings.json), update the connection string to point to your SQL Server instance and database name.
- Build the solution (Ctrl+Shift+B) to restore NuGet packages.
- Press F5 to run. Log in with the default admin credentials set during seeding.

7. Security Considerations

Concern	Approach
Password Storage	Passwords are hashed before storage; plain text is never persisted to the DB
Role Enforcement	IsAdmin flag is checked on login; Admin-only windows are not accessible to Cashier accounts
SQL Injection	All DB operations use parameterized queries via ADO.NET SqlCommand
Session Handling	Logout clears the current user session and returns to the login screen
Data Validation	Input fields are validated before any DB operation is performed

8. Project Structure

Folder / File	Description
CafeteriaProject/	Root solution folder
├── Models/	C# entity classes (User, Item, Category, Invoice)
├── DataAccess/	ADO.NET helper classes and SQL queries
├── Views/	WPF UserControls for each screen (XAML + code-behind)
├── MainWindow.xaml	Shell window containing the sidebar and content area
├── LoginWindow.xaml	Standalone login window
└── App.config	Connection string and application settings

9. Sample Data

9.1 Sample Users

Name	Gender	Role	Phone
Ahmed Hassanein	Male	Admin	0123456780
Sarah Jenkins	Female	Cashier	0987653321
Michael Scott	Male	Cashier	01277889900
Yousef	Male	Admin	0123645230

9.2 Sample Menu Items

Item	Price (EGP)	Category
Espresso	45.00	Hot Drinks
Cappuccino	60.00	Hot Drinks
Fresh Orange Juice	55.00	Cold Drinks
Chocolate Croissant	40.00	Bakery
Turkey & Cheese Wrap	85.00	Sandwiches

10. Screenshots

صفحه لتعديل الاصناف
نظام إدارة الأصناف الذي يسمح للمسؤول بتحديد الأسعار وتحديث تفاصيل المخزون وتعيين الأصناف لمجموعات محددة.
Cafeteria Management System
Welcome
Mr.Yousef
Item Details
Chocolate Croissant
40,00
Category
Bakery
Add Item
Edit Item
Delete Item
ItemCode
ItemName
ItemPrice
CatName
16
Espresso
45.00
Hot Drinks
17
Cappuccino
60.00
Hot Drinks
18
Fresh Orange J...
55.00
Cold Drinks
19
Chocolate Croi...
40.00
Bakery
20
Turkey & Chees...
85.00
Sandwiches
Users
Items
Categories
Invoices
LogOut

صفحه لتعديل التصنيفات
واجهة مخصصة لتنظيم القائمة إلى مجموعات منطقية مثل المشروبات الساخنة والباردة والمخبوزات لتسهيل وتسريع عمل الكاشير.
Cafeteria Management System
Welcome
Mr.Yousef
Category Details
Cold Drinks
Juices, sodas, and iced lattes
Add Category
Edit Category
Delete
CatCode
CatName
CatDesc
1
Hot Drinks
Coffee, tea, and herba...
2
Cold Drinks
Juices, sodas, and ice...
3
Bakery
Freshly baked croissa...
4
Sandwiches
Hot and cold panini a...
Users
Items
Categories
Invoices
LogOut

صفحه لإدارة الفواتير
سجل شامل لجميع عمليات البيع، يتضمن القدرة على البحث برقم الطلب أو التاريخ ومراجعة تفاصيل كل فاتورة.
Cafeteria Management System
Welcome
Mr.Yousef
Users
Items
Categories
Invoices
LogOut
Invoices History
Order Number
Filters
Order Number
Date
Price Range
Search
Reset
InvoiceId
UserNam
InvoiceDa
InvoiceAm
1 Sarah... 3/1/2026 105
2 Michae... 3/2/2026 145
3 Sarah... 3/2/2026 60
4 Sarah... 3/2/2026 215
5 Sarah... 3/3/2026 725
ItemName
ItemPrice
itemQunatity
Total
Fresh Orange ... 55.0000 2 110.0000
Chocolate Cr... 40.0000 4 160.0000
Espresso 45.0000 5 225.0000
Cappuccino 60.0000 1 60.0000
Turkey & Che... 85.0000 2 170.0000

صفحة الفواتير ونقطة البيع
واجهة تفاعلية لنقطة البيع تقوم بحساب الإجمالي في الوقت الفعلي وتدير عربة التسوق بشكل ديناميكي.
Cafeteria Management System
Welcome
Mrs.Sarah Jenkins
Billing
LogOut
Billing Details
Fresh Orange Juice
2
55.00
Remove Item
Add to Cart
Clear Cart
Cart
ItemName
Price
Quantity
Total
Cappuccino 60 3 180
Turkey & Cheese Wrap 85 2 170
Chocolate Croissant 40 4 160
Fresh Orange Juice 55 2 110
Categories List
CatName
Bakery
Cold Drinks
Hot Drinks
Sandwiches
Items List
ItemName
ItemPr
Fresh Orange Juice 55.00
Total = 620.00
Complete Order

11. Contact & Links

Resource	Link
GitHub Repository	https://github.com/joewaleed/Cafeteria_Project
Developer Portfolio	https://yousefwaleed.framer.website
Upwork Profile	Yousef Waleed - C# / .NET / Unity Developer

End of Documentation